Jiarui Hai

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LinkedIn | GitHub | Google Scholar

Research Interest

My research focuses on generative modeling for audio, speech, and music, such as text-to-audio generation, speech synthesis, voice generation, and generative source separation. I also work on audio understanding, including sound event detection, audio language models, and audio information retrieval. Drawing on my background in music production, I bring perspectives on aesthetics, creative practice, and human–system interaction to my research.

Education

Johns Hopkins University — Baltimore, MD, USA
Ph.D Student in Electrical and Computer Engineering — Aug 2022 - Present
Advisor: Prof. Mounya Elhilali

Tsinghua University — Beijing, China
M.E. in Civil Engineering — Aug 2020 - Jun 2022
Member in Big Data Program

Tsinghua University — Beijing, China
B.E. in Civil Engineering — Aug 2016 - Jun 2020
B.S. in Business Analytics

Experience

Johns Hopkins University — Baltimore, MD, USA
Research Assistant, Supervisor: Mounya Elhilali — Aug 2022 - Present
Research Topic: Audio Generation and Understanding

Apple — USA
ML Research Intern — Apr 2026 - Aug 2026
Research Topic: Audio Generative Models (details pending release)

Adobe Research — San Francisco, CA, USA
Research Scientist Intern, Supervisors: Ke Chen, Rithesh Kumar, Zeyu Jin — May 2025 - Feb 2026
Research Topic: Speech Synthesis, Voice Design

Tencent AI Lab — Bellevue, WA, USA
Research Scientist Intern, Supervisors: Yong Xu, Hao Zhang, Dong Yu — May 2024 - Aug 2024
Research Topic: Text-to-Audio Generation, Diffusion Models

Kuaishou AI Platform — Beijing, China
Algorithm Engineer Intern, Supervisors: Rui Lu, Zhiyao Duan — Jun 2020 - Jan 2021
Research Topic: Music Tagging, Singing Voice Synthesis

Selected Publications

Audio · Speech / Voice · Music · Generation · Understanding

Full list available on Google Scholar

* Equal Contribution † Corresponding Author

[19] **Jiarui Hai**, Karan Thakkar, Ke Chen, Yunyun Wang, Jiaqi Su, Rithesh Kumar, Mounya Elhilali, Zeyu Jin. "VoiceDesigner: Text-to-Voice Generation and Editing via Unified Diffusion Modeling and Data Augmentation". Pre-print, 2026. [PDF] [Homepage]

[18] Aaron Yee, Fengjie Lu, **Jiarui Hai**†, Chenang Jiang, Helin Wang, Siwei Tu, Weitao You, Lingyun Sun. "VoiceTrace: A Benchmark and Retrieval Framework for Who-Said-What Speech Retrieval". Pre-print, 2026. [PDF] [Homepage]

[17] Fengjie Lu, Chenang Jiang, **Jiarui Hai**†, Helin Wang, Aaron Yee. "ALM2Vec: Learning Audio Embeddings for Universal Audio Retrieval with Large Audio-Language Models". Technical Report, 2026. [PDF] [Homepage]

[16] Xilin Jiang*, Qiaolin Wang*, Junkai Wu*, Xiaomin He*, Zhongweiyang Xu*, . . ., **Jiarui Hai**, *et al.* "AVMeme Exam: A Multimodal Multilingual Multicultural Benchmark for LLMs' Contextual and Cultural Knowledge and Thinking". COLM 2026. [PDF] [Homepage]

[15] Yongyi Zang, **Jiarui Hai**, Wanying Ge, Qiuqiang Kong, Zheqi Dai, Helin Wang, Yuki Mitsufuji, Mark D. Plumbley. "Summary of The Inaugural Music Source Restoration Challenge". IEEE ICASSP 2026. [PDF] [Homepage]

[14] Helin Wang*, **Jiarui Hai***, Dading Chong, Karan Thakkar, Tiantian Feng, Dongchao Yang, Junhyeok Lee, Laureano Moro Velazquez, Jesus Villalba, Zengyi Qin, Shrikanth Narayanan, Mounya Elhilali, Najim Dehak. "CapSpeech: Enabling Downstream Applications in Style-Captioned Text-to-Speech". IEEE TASLP 2026. [PDF] [Homepage]

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[13] Helin Wang, **Jiarui Hai**, Dongchao Yang, Chen Chen, Kai Li, Junyi Peng, Thomas Thebaud, Laureano Moro-Velazquez, Jesus Villalba, Najim Dehak. “SoloSpeech: Enhancing Intelligibility and Quality in Target Speech Extraction through a Cascaded Generative Pipeline". IEEE TASLP 2026. [PDF] [Homepage]

[12] **Jiarui Hai**, Helin Wang, Weizhe Guo, Mounya Elhilali. “FlexSED: Towards Open-Vocabulary Sound Event Detection". IEEE WASPAA 2025 (**Oral**). [PDF] [Homepage]

[11] **Jiarui Hai**, Mounya Elhilali. “SynSonic: Augmenting Sound Event Detection through Text-to-Audio Diffusion ControlNet and Effective Sample Filtering". IEEE WASPAA 2025. [PDF] [Homepage]

[10] **Jiarui Hai**, Yong Xu, Hao Zhang, Chenxing Li, Helin Wang, Mounya Elhilali, Dong Yu. “EzAudio: Enhancing Text-to-Audio Generation with Efficient Diffusion Transformer". Interspeech 2025 (**Oral**). [PDF] [Homepage]

[9] Helin Wang*, **Jiarui Hai***, Yen-Ju Lu, Karan Thakkar, Mounya Elhilali, Najim Dehak. “SoloAudio: Target Sound Extraction with Language-oriented Audio Diffusion Transformer". IEEE ICASSP 2025. [PDF] [Homepage]

[8] Helin Wang, Meng Yu, **Jiarui Hai**, Chen Chen, Yuchen Hu, Rilin Chen, Najim Dehak, Dong Yu. “SSR-Speech: Towards Stable, Safe and Robust Zero-shot Text-based Speech Editing and Synthesis". IEEE ICASSP 2025. [PDF] [Homepage]

[7] **Jiarui Hai***, Karan Thakkar*, Helin Wang, Zengyi Qin, Mounya Elhilali. “DreamVoice: Text-Guided Voice Conversion". Interspeech 2024. [PDF] [Homepage]

[6] Helin Wang, Jesus Villalba, Laureano Moro-Velazquez, **Jiarui Hai**, Thomas Thebaud, Najim Dehak. “Noise-robust Speech Separation with Fast Generative Correction". Interspeech 2024 (**Best Paper Award**). [PDF]

[5] **Jiarui Hai***, Helin Wang*, Dongchao Yang, Karan Thakkar, Najim Dehak, Mounya Elhilali. “DPM-TSE: A Diffusion Probabilistic Model for Target Sound Extraction". IEEE ICASSP 2024. [PDF] [Homepage]

[4] Karan Thakkar, **Jiarui Hai**, Mounya Elhilali. “Investigating Self-Supervised Deep Representations for EEG-Based Auditory Attention Decoding". IEEE ICASSP 2024. [PDF]

[3] **Jiarui Hai***, Yu-jeh Liu*, Mounya Elhilali. “Boosting Modality Representation with Pre-trained Models and Multi-task Training for Multimodal Sentiment Analysis". IEEE ASRU 2023. [PDF]

[2] **Jiarui Hai**, Mounya Elhilali. “Diff-Pitcher: Diffusion-based Singing Voice Pitch Correction". IEEE WASPAA 2023 (**Oral**). [PDF] [Homepage]

[1] Rui Lu, Baigong Zheng, **Jiarui Hai**, Fei Tao, Zhiyao Duan, Ji Liu. “Progressive Teacher-Student Training Framework for Music Tagging". IEEE ICASSP 2022. [PDF]

Open-source Resources

Datasets, Benchmarks, and Repositories
- Music Source Restoration Benchmark: **[Homepage]**
- CapSpeech: A large-scale voice captioning dataset: **[Homepage]**
- DreamVoice: A voice timbre description dataset: **[Homepage]**

Open-Source Repositories
- OpenSound: Built interactive demos for audio and speech models on Hugging Face: **[Hugging Face]**
- MeanFlow: PyTorch implementation of MeanFlow for diffusion accelerating: **[Code]**
- EzAudio: Efficient diffusion transformer optimized for audio latents: **[Code]**

Service and Teaching

Peer Reviewer
- Conference: ICASSP, Interspeech, WASPAA, ASRU, SLT
- Journal: IJCV, TASLP, JBHI

Challenge Organizer
- ICASSP 2026 The Inaugural Music Source Restoration Challenge: **[Homepage]**

Teaching Assistant
- Audio Signal Processing (EN.520.445/645), Johns Hopkins University, Fall 2025
- Audio Signal Processing (EN.520.445/645), Johns Hopkins University, Fall 2024

Music Activities

Professional Recognition
- Featured Producer on a rap single showcased in HIPHOP BANK (TV show, China) Aug 2021
- Invited Lecture on Music Production, Modern Sky (Chinas leading music company) Jun 2021
- Top New Producer Award, BeatsHome Hip-hop Production Contest Apr 2021

Creative & Media Activities
- Independent Artist & Beatmaker: produce singles and beat tapes
- Content Creator: share tutorials on music production and mixing

Leadership and Co-founder
- President, Rap & HipHop Association, Tsinghua University 2020–2021
- Co-founder, InnerSup Records (Music Producer Team)
- Co-founder, Music Production Team, Rap & HipHop Association, Tsinghua University

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